

- The straight lines l_1 and l_2 pass through the origin and trisect the line segment of the line $L : 9x + 5y = 45$ between the axes. If m_1 and m_2 are the slopes of the lines l_1 and l_2 , then the point of intersection of the line $y = (m_1 + m_2)x$ with L lies on
[2023 (06 Apr Shift 1)]
(1) $y - 2x = 5$
(2) $6x + y = 10$
(3) $y - x = 5$
(4) $6x - y = 15$
- Let $A(0, 1)$, $B(1, 1)$ and $C(1, 0)$ be the mid-points of the sides of a triangle with incentre at the point D . If the focus of the parabola $y^2 = 4ax$ passing through D is $(\alpha + \beta\sqrt{2}, 0)$, where α and β are rational numbers, then $\frac{\alpha}{\beta^2}$ is equal to
[2023 (08 Apr Shift 2)]
(1) 8
(2) 12
(3) 6
(4) $\frac{9}{2}$
- The area of the quadrilateral $ABCD$ with vertices $A(2, 1, 1)$, $B(1, 2, 5)$, $C(-2, -3, 5)$ and $D(1, -6, -7)$ is equal to
[2023 (08 Apr Shift 2)]
(1) 48
(2) $8\sqrt{38}$
(3) 54
(4) $9\sqrt{38}$
- A line segment AB of length λ moves such that the points A and B remain on the periphery of a circle of radius λ . Then the locus of the point, that divides the line segment AB in the ratio 2 : 3, is a circle of radius
[2023 (10 Apr Shift 1)]
(1) $\frac{3}{5}\lambda$
(2) $\frac{2}{3}\lambda$
(3) $\frac{\sqrt{19}}{5}\lambda$
(4) $\frac{\sqrt{19}}{7}\lambda$
- Let A be the point $(1, 2)$ and B be any point on the curve $x^2 + y^2 = 16$. If the centre of the locus of the point P , which divides the line segment AB in the ratio 3 : 2 is the point $C(\alpha, \beta)$, then the length of the line segment AC is
[2023 (10 Apr Shift 2)]
(1) $\frac{3\sqrt{5}}{5}$
(2) $\frac{4\sqrt{5}}{5}$
(3) $\frac{2\sqrt{5}}{5}$
(4) $\frac{6\sqrt{5}}{5}$
- Let the equations of two adjacent sides of a parallelogram $ABCD$ be $2x - 3y = -23$ and $5x + 4y = 23$. If the equation of its one diagonal AC is $3x + 7y = 23$ and the distance of A from the other diagonal is d , then $50d^2$ is equal to _____
[2023 (10 Apr Shift 2)]
- Let R be a rectangle given by the lines $x = 0$, $x = 2$, $y = 0$ and $y = 5$. Let $A(\alpha, 0)$ and $B(0, \beta)$, $\alpha \in [0, 2]$ and $\beta \in [0, 5]$, be such that the line segment AB divides the area of the rectangle R in the ratio 4 : 1. Then, the mid-point of AB lies on a
[2023 (11 Apr Shift 1)]
(1) straight line
(2) parabola
(3) hyperbola
(4) circle
- If the line $l_1 : 3y - 2x = 3$ is the angular bisector of the lines $l_2 : x - y + 1 = 0$ and $l_3 : \alpha x + \beta y + 17 = 0$, then $\alpha^2 + \beta^2 - \alpha - \beta$ is equal to
[2023 (11 Apr Shift 2)]
- If the point $(\alpha, \frac{7\sqrt{3}}{3})$ lies on the curve traced by the mid-points of the line segments of the lines $x \cos \theta + y \sin \theta = 7$, $\theta \in (0, \frac{\pi}{2})$ between the co-ordinates axes, then α is equal to
[2023 (12 Apr Shift 1)]
(1) -7
(2) $-7\sqrt{3}$
(3) $7\sqrt{3}$
(4) 7

10. Let (α, β) be the centroid of the triangle formed by the lines $15x - y = 82$, $6x - 5y = -4$ and $9x + 4y = 17$. Then $\alpha + 2\beta$ and $2\alpha - \beta$ are the roots of the equation

[2023 (13 Apr Shift 2)]

(1) $x^2 - 7x + 12 = 0$

(2) $x^2 - 14x + 48 = 0$

(3) $x^2 - 13x + 42 = 0$

(4) $x^2 - 10x + 25 = 0$

11. If (α, β) is the orthocenter of the triangle ABC with vertices $A(3, -7)$, $B(-1, 2)$ and $C(4, 5)$, then $9\alpha - 6\beta + 60$ is equal to

[2023 (15 Apr Shift 1)]

(1) 25

(2) 35

(3) 30

(4) 40

ANSWER KEYS

1. (3) 2. (1) 3. (2) 4. (3) 5. (1) 6. (529) 7. (3) 8. (348)
9. (4) 10. (3) 11. (1)

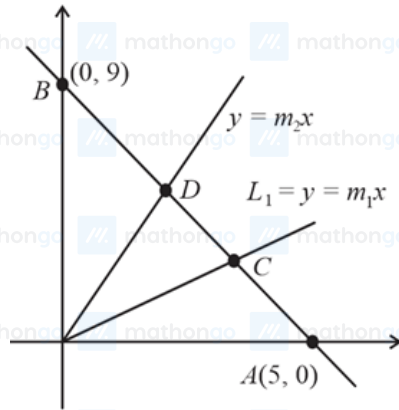
1. (3)

Given,

The straight lines l_1 and l_2 pass through the origin and trisect the line segment of the line $L : 9x + 5y = 45$ between the axes,

And m_1 and m_2 are the slopes of the lines l_1 and l_2 ,

Now on plotting the diagram we get,



Given equation of line, $L : 9x + 5y = 45$

$$\Rightarrow \frac{x}{5} + \frac{y}{9} = 1$$

Now using the section formula between point $A(5, 0)$ & $B(0, 9)$ we get the value of point C & D

$$\Rightarrow C \equiv \left(\frac{10}{3}, 3\right) \text{ and } D \equiv \left(\frac{5}{3}, 6\right)$$

Now finding the slope m_1 & m_2 we get,

$$m_1 = \frac{3-0}{\frac{10}{3}-0} = \frac{9}{10} \text{ \& } m_2 = \frac{6-0}{\frac{5}{3}-0} = \frac{18}{5}$$

So, equation of line $y = (m_1 + m_2)x$ will be,

$$y = \left(\frac{9}{10} + \frac{36}{10}\right)x = \frac{9}{2}x$$

So, intersection point with L will be,

$$7y = 45 \Rightarrow y = \frac{45}{7}, x = \frac{10}{7}$$

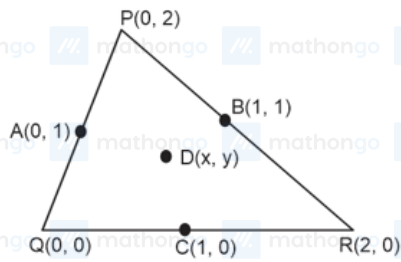
$$\text{Hence, } y - x = \frac{45-10}{7} = 5$$

2. (1)

Given,

$A(0, 1)$, $B(1, 1)$ and $C(1, 0)$ be the mid-points of the sides of a triangle with incentre at the point D ,

Now finding the vertices of the triangle by using midpoint formula and plotting the diagram we get,



Now finding the incentre of the above triangle using the formula,

$$I \equiv \left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c} \right)$$

$$\text{We get, Incentre } D = \left(\frac{4}{4+2\sqrt{2}}, \frac{4}{4+2\sqrt{2}} \right)$$

Now given parabola $y^2 = 4ax$ passes through the incentre D we get,

$$\left(\frac{4}{4+2\sqrt{2}} \right)^2 = 4a \left(\frac{4}{4+2\sqrt{2}} \right)$$

$$\Rightarrow a = \frac{1}{4+2\sqrt{2}}$$

Now we know that focus of parabola is given by, focus $(a, 0)$

$$\text{Hence, focus will be, } \left(\frac{1}{4+2\sqrt{2}}, 0 \right) = \left(\frac{4-2\sqrt{2}}{8}, 0 \right)$$

Now comparing with $(\alpha + \beta\sqrt{2}, 0)$ we get,

$$\alpha = \frac{4}{8} = \frac{1}{2}, \quad \beta = -\frac{1}{4}$$

$$\text{Hence, } \frac{\alpha}{\beta^2} = \frac{\frac{1}{2}}{\left(-\frac{1}{4}\right)^2} = \frac{16}{2} = 8$$

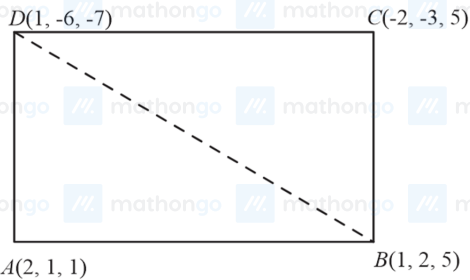
3. (2)

Given,

The quadrilateral $ABCD$ with vertices $A(2, 1, 1)$, $B(1, 2, 5)$, $C(-2, -3, 5)$ and $D(1, -6, -7)$

Now finding, $\vec{AB} = -\hat{i} + \hat{j} + 4\hat{k}$ and $\vec{AD} = -\hat{i} - 7\hat{j} - 8\hat{k}$

And plotting the diagram we get,



Now we know that,

Area of $\triangle ABD$ is given by,

$$= \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 4 \\ -1 & -7 & 8 \end{vmatrix}$$

$$= |10\hat{i} - 6\hat{j} + 4\hat{k}| = 2\sqrt{38}$$

Now finding, $\vec{CB} = 3\hat{i} + 5\hat{j}$ and $\vec{CD} = 3\hat{i} - 3\hat{j} - 12\hat{k}$

$$\text{So, Area of } \triangle CBD = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -3 & -12 \\ 3 & 5 & 0 \end{vmatrix}$$

$$= |6(5\hat{i} - 3\hat{j} - 2\hat{k})|$$

$$= 6\sqrt{38}$$

$\therefore$ Area of quadrilateral $ABCD = 8\sqrt{38}$ square units

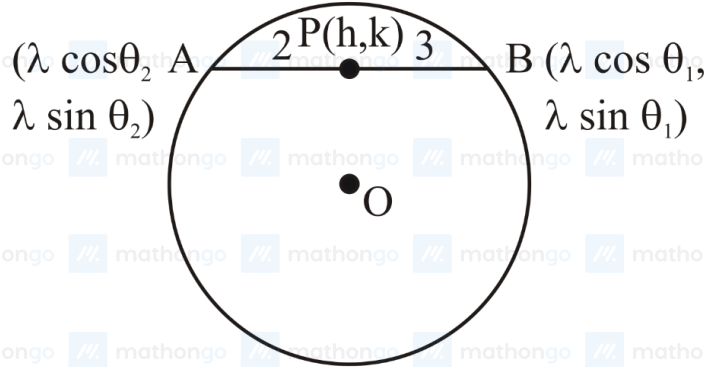
4. (3)

Given,

A line segment AB of length λ moves such that the points A and B remain on the periphery of a circle of radius λ .

Now taking the points on the circle of radius λ as $B(\lambda \cos \theta_1, \lambda \sin \theta_1)$ & $A(\lambda \cos \theta_2, \lambda \sin \theta_2)$ and taking the point $P(h, k)$ which divides the line segment in AB of length λ in $2 : 3$

Now plotting the diagram we get,



Now, let O be the origin and radius of circle is λ and $AB = \lambda$ and using distance formula we get,

$$AB = \lambda = \sqrt{(\lambda \cos \theta_1 - \lambda \cos \theta_2)^2 + (\lambda \sin \theta_1 - \lambda \sin \theta_2)^2}$$

$$\Rightarrow 1 = 2 - 2 \cos(\theta_1 - \theta_2)$$

$$\Rightarrow \cos(\theta_1 - \theta_2) = \frac{1}{2}$$

Now using section formula we get,

$$h = \frac{2\lambda \cos \theta_1 + 3\lambda \cos \theta_2}{5} \text{ and } k = \frac{2\lambda \sin \theta_1 + 3\lambda \sin \theta_2}{5}$$

Now squaring and adding above two value we get,

$$h^2 + k^2 = \frac{\lambda^2}{25} [4 + 9 + 12(\cos(\theta_1 - \theta_2))]$$

$$\Rightarrow h^2 + k^2 = \frac{\lambda^2}{25} \cdot 19$$

$$\text{Hence, Radius} = \frac{\lambda}{5} \sqrt{19}$$

5. (1)

Given,

A be the point $(1, 2)$ and B be any point on the curve $x^2 + y^2 = 16$,

And the centre of the locus of the point P , which divides the line segment AB in the ratio $3 : 2$ is the point $C(\alpha, \beta)$,

Now let the point on the circle $x^2 + y^2 = 16$ be $B(4 \cos \theta, 4 \sin \theta)$

Now using section formula in $A(1, 2)$ and $B(4 \cos \theta, 4 \sin \theta)$ we get,

$$P\left(\frac{12 \cos \theta + 2}{5}, \frac{12 \sin \theta + 4}{5}\right) \equiv (h, k)$$

$$\Rightarrow \cos \theta = \frac{5h-2}{12} \text{ \& \; } \sin \theta = \frac{5k-4}{12}$$

Now squaring and adding we get,

$$\left(\frac{5h-2}{12}\right)^2 + \left(\frac{5k-4}{12}\right)^2 = 1$$

$$\Rightarrow \left(h - \frac{2}{5}\right)^2 + \left(k - \frac{4}{5}\right)^2 = \left(\frac{12}{5}\right)^2$$

$$\text{So, the centre of the locus is } C\left(\frac{2}{5}, \frac{4}{5}\right)$$

$$\text{Hence, by distance formula we get, } AC = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{6}{5}\right)^2} = \frac{3\sqrt{5}}{5}$$

6. (529)

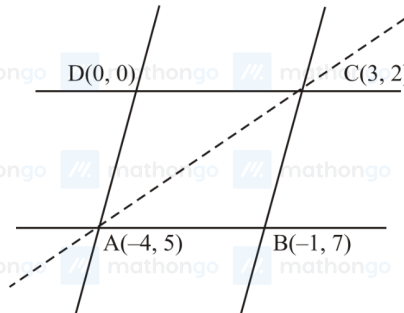
Given,

The equations of two adjacent sides of a parallelogram $ABCD$ be $2x - 3y = -23$ and $5x + 4y = 23$,

So, $AB \equiv 2x - 3y = -23$ and $BC \equiv 5x + 4y = 23$

Also given, $AC \equiv 3x + 7y = 23$

Solving the above lines we get, $A(-4, 5)$, $B(-1, 7)$, $C(3, 2)$



We know that,

Diagonal of parallelogram have same midpoint,

So AC and BD have same mid-point and let point D be (x, y) ,

So midpoint formula we get,

$$\frac{x-1}{2} = \frac{-4+3}{2} \Rightarrow x = 0 \text{ and } \frac{y+7}{2} = \frac{2+5}{2} \Rightarrow y = 0$$

Hence, point D is $(0, 0)$

Now Equation of BD will be $7x + y = 0$

Now finding the distance of $A(-4, 5)$ from $7x + y = 0$ we get,

$$d = \left| \frac{7(-4)+5}{\sqrt{7^2+1^2}} \right| = \frac{23}{\sqrt{50}}$$

$$\text{Hence, } 50d^2 = 23^2 = 529$$

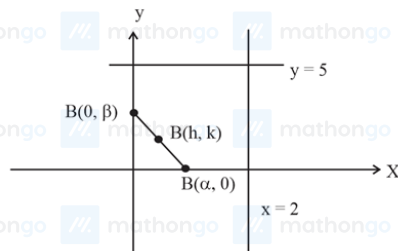
7. (3)

Given,

R be a rectangle given by the lines $x = 0$, $x = 2$, $y = 0$ and $y = 5$

And point $A(\alpha, 0)$ and $B(0, \beta)$, $\alpha \in [0, 2]$ and $\beta \in [0, 5]$,

Now plotting the diagram we get,



Now given that line segment AB divides the ratio of area in $4 : 1$, we get

$$\frac{10 - \frac{1}{2}\alpha\beta}{\frac{1}{2}\alpha\beta} = \frac{4}{1}$$

$$\Rightarrow 20 - \alpha\beta = 4\alpha\beta$$

$$\Rightarrow \alpha\beta = 4 \dots \dots (1)$$

Now using midpoint formula we get,

$$h = \frac{\alpha}{2} \text{ \& } \beta = \frac{k}{2}$$

Now using equation (1) we get,

$$\Rightarrow 4hk = 4$$

$\Rightarrow xy = 1$ which is a equation of rectangular hyperbola.

8. (348)

Given,

$L_1 : 3y - 2x = 3$ is angular bisector of $L_2 : x - y + 1 = 0$ and $L_3 : \alpha x + \beta y + 17 = 0$

Now finding point of intersection of L_1 & L_2 we get, $(0, 1)$

And point will lie on L_3 , so $\alpha \times 0 + \beta \times 1 + 17 = 0$

$$\Rightarrow \beta = -17$$

Any point, say $\left(\frac{-3}{2}, 0\right)$ on L_1 should be equidistant from lines L_2 & L_3

Now using the formula of distance of a point from the line we get,

$$\Rightarrow \left| \frac{\frac{-3}{2} - 0 + 1}{\sqrt{1^2 + 1^2}} \right| = \left| \frac{\frac{-3\alpha}{2} + 0 + 17}{\sqrt{\alpha^2 + (-17)^2}} \right|$$

$$\Rightarrow (\alpha - 7)(\alpha - 17) = 0$$

Now, for $\alpha = 17$, L_2 & L_3 coincides

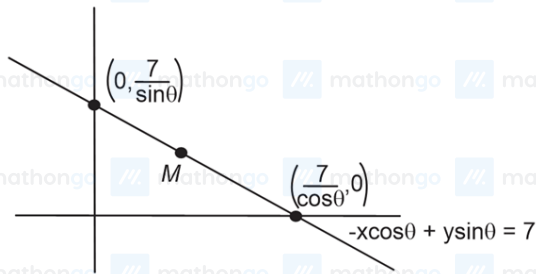
So, $\alpha = 7$

Now putting the value of α & β in given expression we get,

$$\alpha^2 + \beta^2 - \alpha - \beta = (-17)^2 + 7^2 - 7 + 17 = 348$$

9. (4)

The straight line is shown below with the intercepts.



$$M\left(\frac{7}{2\cos\theta}, \frac{7}{2\sin\theta}\right) \equiv (h, k)$$

$$\therefore \cos\theta = \frac{7}{2h}, \sin\theta = \frac{7}{2k}$$

$$\therefore \theta \in \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow \frac{49}{4h^2} + \frac{49}{4k^2} = 1 \quad \therefore x \text{ and } y \text{ are positive}$$

$$\therefore \text{Locus} = \frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{49}$$

Now $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ lies on curve

$$\Rightarrow \frac{1}{\alpha^2} + \frac{9}{147} = \frac{4}{49}$$

$$\frac{1}{\alpha^2} = \frac{1}{49}$$

$$\therefore \alpha = \pm 7$$

$$\therefore \alpha = 7$$

Hence this is the required option.

10. (3)

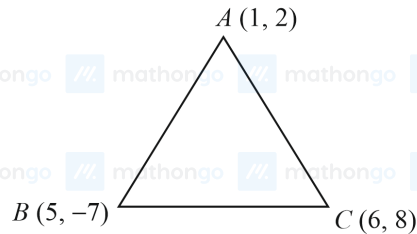
The given set of equations are:

$$15x - y = 82$$

$$6x - 5y = -4$$

$$9x + 4y = 17$$

Now on solving the above equation by taking two at a time and plotting the diagram we get,



$$(\alpha, \beta) = \left(\frac{1+5+6}{3}, \frac{2-7+8}{3} \right) = (4, 1)$$

Hence, $\alpha + 2\beta = 6$ and $2\alpha - \beta = 7$

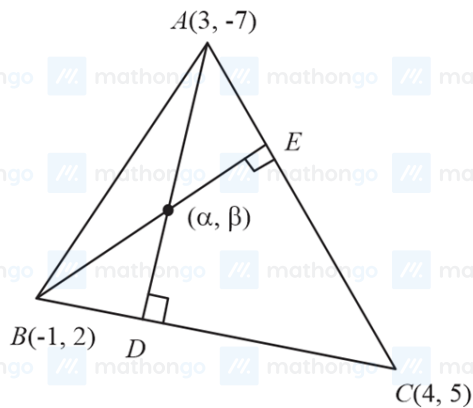
The required equation is $(x - 6)(x - 7) = 0$

$$\text{Equation } x^2 - 13x + 42 = 0$$

Hence this is the required option.

11. (1)

The required diagram will be:



$$\text{Equation of } AD : y + 7 = \frac{-5}{3}(x - 3)$$

$$\Rightarrow 3y + 21 = -5x + 15$$

$$\Rightarrow 5x + 3y + 6 = 0 \dots\dots(i)$$

$$\text{Equation of } BE : y - 2 = \frac{-1}{12}(x + 1)$$

$$\Rightarrow 12y - 24 = -x - 1$$

$$\Rightarrow x = 23 - 12y \dots(ii)$$

$$\text{by (ii) } 115 - 60y + 3y + 6 = 0$$

$$\Rightarrow 57y = 121$$

$$y = \frac{121}{57}, x = 23 - 12 \times \frac{121}{57}$$

$$\therefore 9\alpha - 6\beta + 60 = 9 \times 23 - 108 \times \frac{121}{57} + 6 \times \frac{121}{57} + 60$$

$$= 207 - 242 + 60 = 25$$

Hence this is the required option.